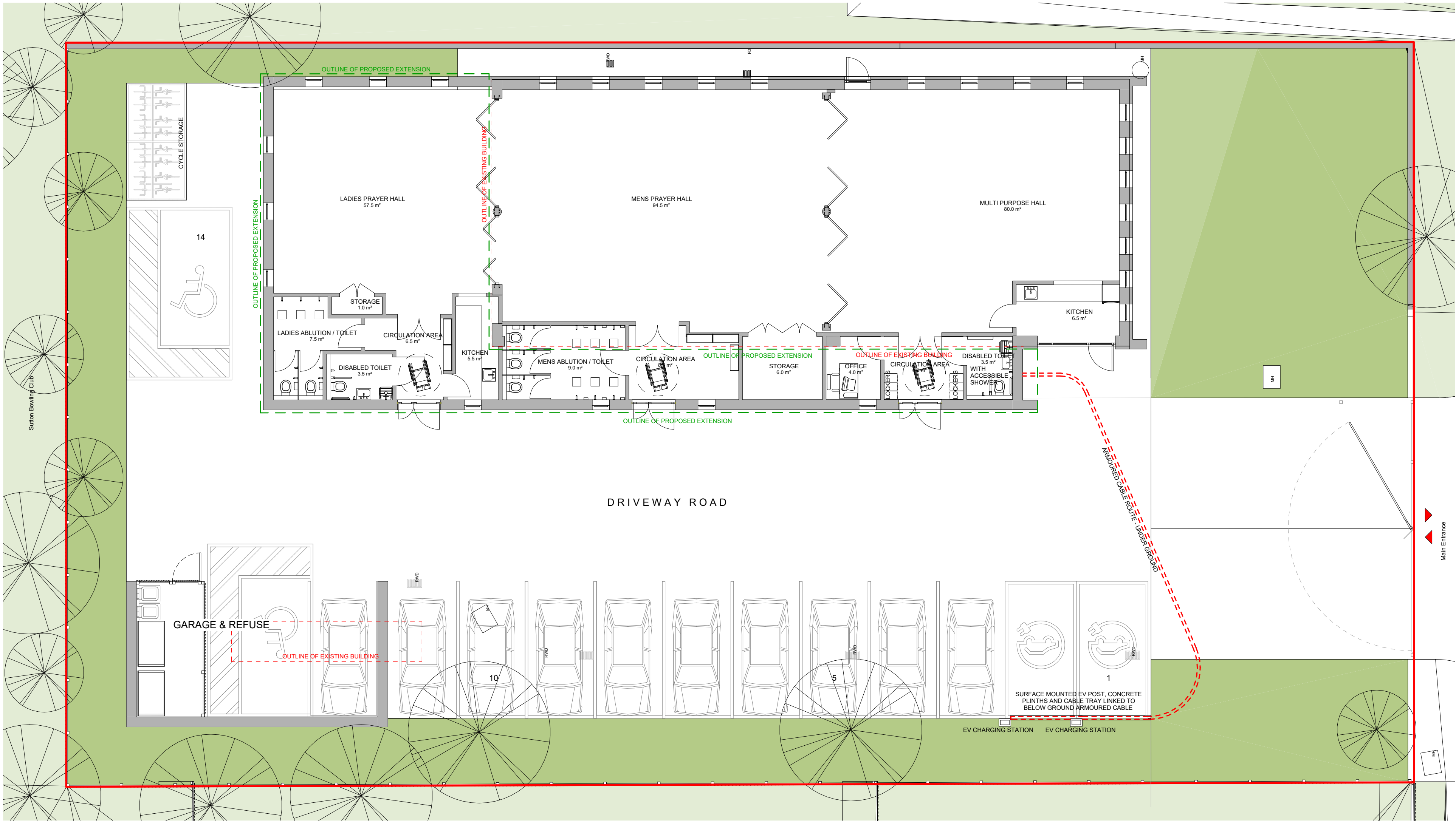




1 Proposed Site Layout Plan  
1 : 100  
Total car park space - 12 + 2 disabled

-Part A relates to Arboricultural and Root Protection Compliance  
-Part B relates to our interaction with installers and specifically EV Charging Installation Compliance  
-Part C is our Compliance Statement Summary

Part A – Our Arboricultural and Root Protection Compliance Method

1. Overview

This review confirms that the proposed installation of two EV charging points and their associated infrastructure fully complies with the **no-dig** and **root-protection** measures identified on **Arbor Cultural Ltd Drawing AC.2025.861– TPP-01 Rev A**, prepared in accordance with **BS 5837:2012 – Trees in Relation to Design, Demolition and Construction– Recommendations**.  
The works have been designed to avoid any excavation or ground disturbance within the defined **Root Protection Areas (RPAs)** of retained trees.

2. Summary of Proposed Methodology

- Install **two EV charging posts** adjacent to the existing car-parking area.
- Construct **100 mm permeable surface build-up** (e.g. open-graded sub-base + porous finish) **above** existing levels – no excavation or compaction of subsoil.
- Route **SWA/Armourflex** cabling from the building internally, exiting externally **outside the RPA** within **HDPE ducting** beneath existing tarmac.
- At the edge of the RPA, the cable **emerges above ground** the onward connection to EV posts is via **surface-mounted cable tray** (galvanised or aluminium).
- Each EV post fixed to a **surface-mounted reinforced concrete pad** (approx. 900 × 500 × 500 mm high), **resting on** the no-dig surface rather than penetrating it.
- All works are **fully reversible** and **installed from existing hardstanding** – no excavation, no machinery within RPA.

3. Compliance Matrix

| Element                       | Relevant BS 5837 / TPP Requirement                                                       | Compliance Assessment | Notes / Actions                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------|
| New surfacing above RPA       | No-dig, laid over existing grade using permeable granular or cellular confinement layer. | Compliant             | 100 mm build-up preserves root aeration. Use geotextile + angular stone (4/20 mm).                         |
| Cable routing                 | Services to avoid RPAs where possible.                                                   | Compliant             | Cable remains outside RPA in duct; above-ground only at transition. Surface tray – no excavation required. |
| Service connection within RPA | Only above-ground or air-spaced under supervision                                        | Compliant             |                                                                                                            |
| EV post bases                 | No excavation; lightweight or surface-mounted only.                                      | Compliant             | Pads rest on surface; consider precast blocks if required.                                                 |
| Soil compaction               | None permitted within CEZ / RPA                                                          | Compliant             | All works from existing hardstand.                                                                         |
| Arboricultural supervision    | Monitor works adjacent RPA                                                               | Recommended           | Site attendance at initial setup and post-installation verification.                                       |

4. Summary

Our methodology:  
-Prevents root severance or compaction  
-Preserves soil permeability and gaseous exchange  
-Avoids any intrusive foundations or excavation  
-Maintains full reversibility of works.

5. Conclusion:

The proposed EV charging installation is **fully compliant** with the arboricultural no-dig requirements and BS 5837:2012. Subject to arborist supervision during initial setting-out, the design presents **no risk to retained tree roots**.

Below is **Part B – EV Charging Installation Compliance Review** based on our proposal **two above-ground EV charging posts**, **surface-mounted containment**, **100+ mm permeable build-up** over no-dig zone, **armoured cable** from building within **HDPE duct**, etc.

This table cross-references each design element against the **relevant UK standards, codes of practice, and Building Regulations**, showing compliance status and concise professional notes.

Part B -EV Charging Installation Compliance Matrix

| Design Element / Aspect                       | Relevant Standard / Regulation                                                         | Compliance Review & Commentary                                                                                                                                                                                                                                                                                                                           | Status                                     |
|-----------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| General electrical installation for EV points | BS 7671:2018 +A2:2022 (IET Wiring Regulations) – Section 722                           | All fixed wiring, protective devices, and earthing arrangements must comply with Part 7 (special installations for EV). Proposed use of SWA/Armourflex cabling and RCD-protected circuit from main distribution board satisfies base requirement. Final cable sizing, RCD type A or B, and maximum voltage drop to be verified by qualified electrician. | Compliant (subject to design verification) |
| Charging equipment type / connector           | BS EN IEC 61851-1:2019 and BS EN IEC 62196-2:2017                                      | Standard UK 'Mode 3 – Type 2' charging points anticipated. Ensure manufacturer certification to BS EN standards.                                                                                                                                                                                                                                         | required                                   |
| Location & accessibility                      | Building Regulations Part S (2021 Ed.) + IET Code of Practice (4th Ed.) Sections 3 & 4 | Two publicly accessible charging bays satisfy minimum provision. Units mounted 500 mm above grade provide appropriate clearance. Cables not obstructing pedestrian routes; clear signage and access zone to be maintained.                                                                                                                               | required                                   |
| Earthing arrangement                          | BS 7671 Sec. 722.411.4 + IET CoP Ch. 7                                                 | PEN (TN-C-S) earthing not to be used unless protective devices or open-PEN protection incorporated. Recommend use of TT Island earthing or PEN fault protection integrated in charger.                                                                                                                                                                   | Check required with installer              |
| Residual Current Device (RCD) protection      | BS EN 50570 / A1:2017 & BS 7671 722.531.3                                              | Each charging circuit must include 30 mA RCD (Type A or B per manufacturer). Dual pole disconnection required.                                                                                                                                                                                                                                           | required                                   |
| Load assessment / capacity                    | IET CoP Ch. 6 and Energy Networks Association G99 / G99                                | Building supply capacity to be checked; apply for DNO notification if total load > 3.6 kW per charger.                                                                                                                                                                                                                                                   | To be verified                             |
| Cable routing & containment                   | IET CoP Sec. 8 & BS 7671 521.10                                                        | HDPE duct below existing tarmac outside RPA, surface-mounted tray above ground across no-dig zone. No buried services within RPA – fully compliant with arboricultural constraints.                                                                                                                                                                      | required                                   |
| Foundations / fixings of EV posts             | IET CoP Sec. 5.2 & BS 8300-1 (accessibility)                                           | Posts surface-mounted on reinforced concrete pads above no-dig area – compliant if posts mechanically anchored (not cast-in). Check manufacturer base-plate loading.                                                                                                                                                                                     | required                                   |
| Environmental exposure & protection           | BS EN 60529 (IP Rating) – minimum IP44 external                                        | Charger units to have IP54 or higher for weather protection. Confirm vandal-resistant housing for public site.                                                                                                                                                                                                                                           | required                                   |
| Signage & bay markings                        | DfT Traffic Signs Regulations 2016 / IET CoP Sec. 10                                   | EV bays to be clearly marked with universal 'EV Charging Only' symbol.                                                                                                                                                                                                                                                                                   | required                                   |
| Lighting & safety                             | BS 5490-1:2020 & IET CoP Sec. 10                                                       | Adequate lighting for night-time use; ensure glare not directed at neighbouring dwellings.                                                                                                                                                                                                                                                               | required                                   |
| Accessibility & Equality                      | BS 8300-1:2018 + A1:2023 – Design for Inclusive Environments                           | Charging posts to be positioned ≤ 1.4 m reach height; bays > 3.6 m wide if accessible use required.                                                                                                                                                                                                                                                      | if adopted for public use                  |
| Energy efficiency / smart control             | Part S Reg. 440 & OZEV guidance 2022                                                   | Chargers to be "smart"-enabled (load balancing, time control). Confirm manufacturer compliance for grant eligibility.                                                                                                                                                                                                                                    | required                                   |

Part B summary

This methodology fully supports compliance with **BS 5837:2012** (no-dig) *and* electrical safety standards.

The key compliance checks still required before final installation are:

- Confirmation of **earthing type / open-PEN protection**
- Verification of **supply capacity** and **DNO notification**
- Installer's certification to **BS 7671** at completion (Electrical Installation Certificate).

Part C -This is our Integrated Compliance Statement:

1. Overview

This integrated compliance statement confirms that the proposed installation of **two surface-mounted EV-charging units** within the car-park area of the **Former Ambulance Station, Belmont (SM2 6JH)** has been designed in full accordance with:

- BS 5837: 2012 – Trees in Relation to Design, Demolition and Construction– Recommendations** **Arbor Cultural Ltd Drawing AC.2025.861 TPP-01 Rev A – Tree Protection Plan & No-Dig Zone**
- BS 7671: 2018 (+A2: 2022) – IET Wiring Regulations, Section 722**
- IET Code of Practice for Electric Vehicle Charging Equipment Installation** (4th Edition) and
- Approved Document S – Infrastructure for Charging Electric Vehicles** (2021 Edition).

The combined methodology ensures both arboricultural protection and statutory electrical safety compliance, preventing disturbance to protected root systems while meeting national standards for EV infrastructure.

2. Arboricultural Compliance Summary

- All works within the Root Protection Area (RPA) follow **no-dig construction methodology**
- A **100 mm permeable build-up** will be laid **above existing ground**, avoiding excavation or compaction.
- Armoured cabling (SWA/Armourflex)** will run **externally to the RPA** in **HDPE ducting** under existing tarmac, emerging **above ground only** where it transitions to surface containment.
- EV-charging posts** will sit on **surface-mounted reinforced-concrete pads** resting on the no-dig layer – no excavation or root disturbance.
- All containment between pads will be **surface-mounted cable tray**, ensuring reversibility and continued soil aeration.
- Installation will occur entirely from existing hardstanding under arboricultural supervision.

Our proposal is designed to fully comply with **BS 5837 Clause 7.2-7.7**, safeguarding retained trees and their root zones.

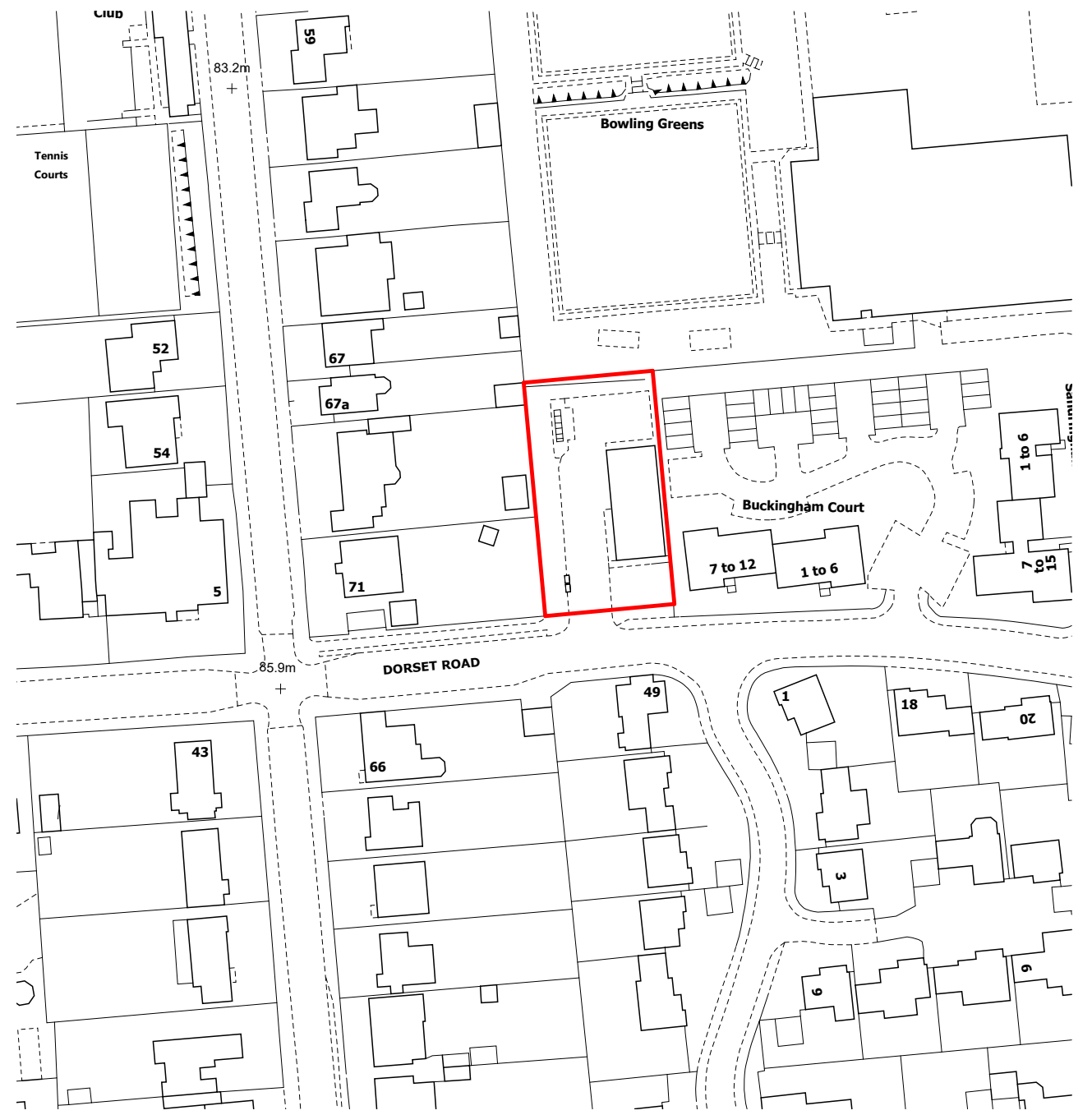
3. EV Charging Standards Compliance Summary

- Electrical Safety:** All circuits will comply with BS 7671 Section 722, including 30 mA RCD protection, Type A/B devices, and suitable open-PEN or TT earthing protection.
- Equipment Certification:** Charging units to conform to BS EN IEC 61851-1 and BS EN 62196-2.
- Accessibility & Location:** Installations meet Part S and BS 8300-1 for access, height, and clear zones.
- Smart Functionality:** Units to include load-management and smart-charging capability per OZEV 2022 guidance.
- Environmental Durability:** All housings minimum IP54 rated; surface fixings to BS EN 60529.

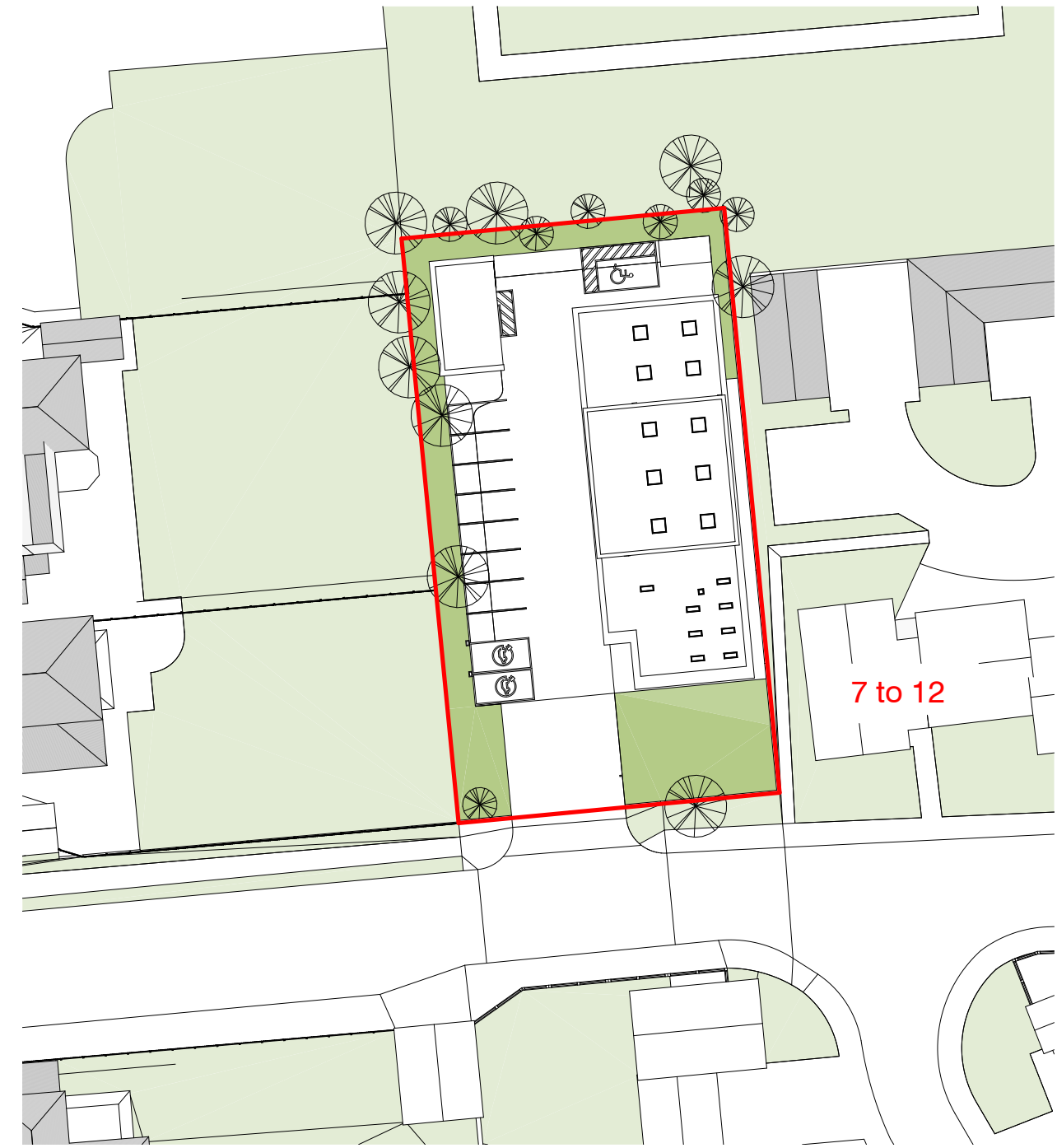
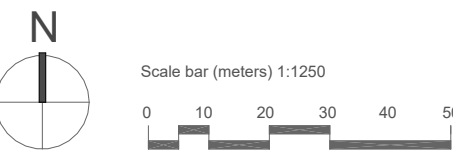
Our proposal satisfies national EV-charging infrastructure regulations and Building Regulations Part S requirements for new or upgraded car-parking facilities.

4. Our Conclusion

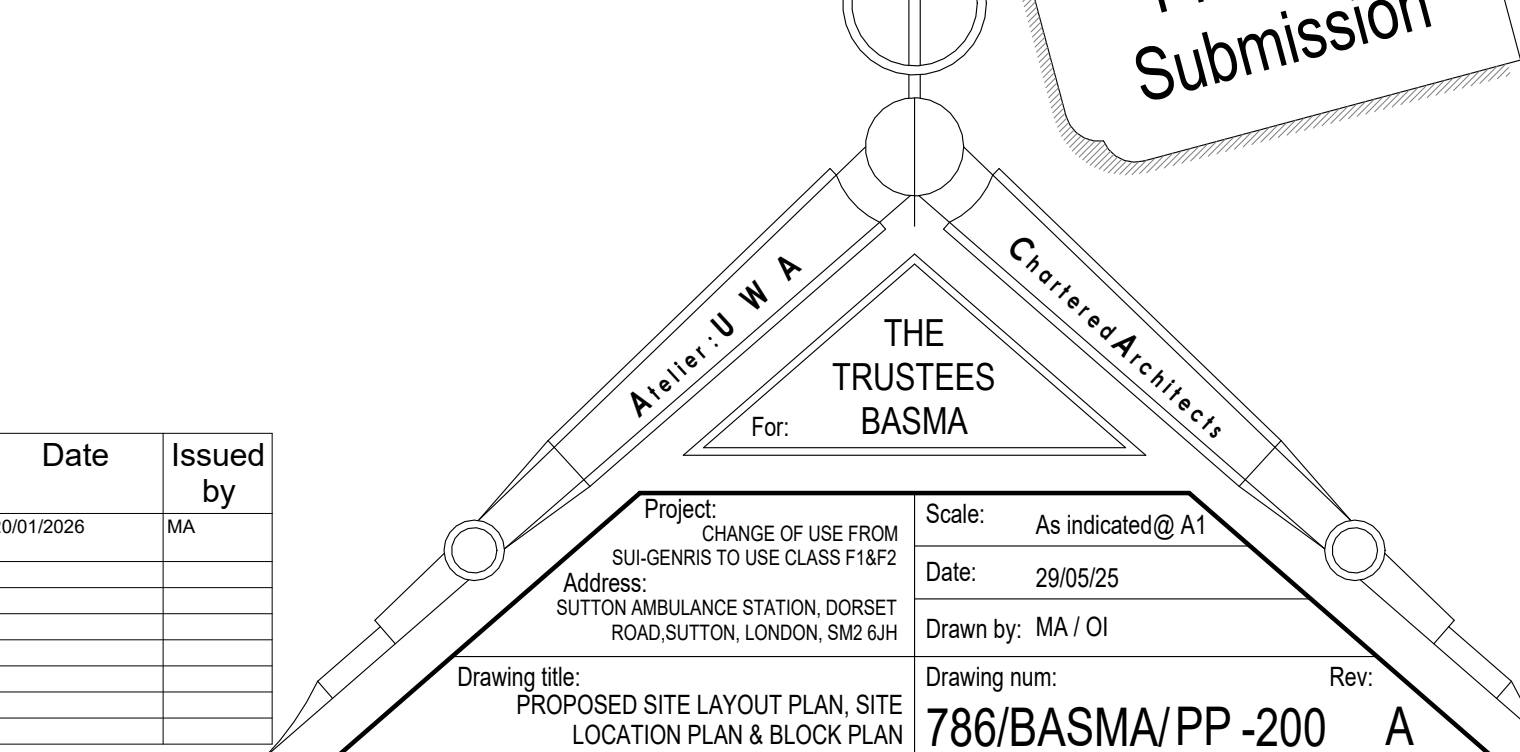
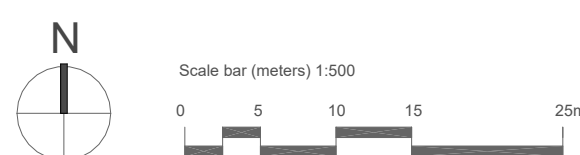
The proposed works achieve dual compliance with both **arboricultural protection** and **EV-charging installation standards**.  
The methodology introduces **no excavation or adverse impact on retained trees** maintains full reversibility, and provides **safe, standards-compliant electrical infrastructure** suitable for Building Control and planning approval.



3 Site Location plan  
1 : 1250



2 Block Plan  
1 : 500



| No. | Description                                                            | Date       | Issued by |
|-----|------------------------------------------------------------------------|------------|-----------|
| A   | EV Charging and Parking location change and additional notes are added | 20/01/2026 | MA        |
|     |                                                                        |            |           |
|     |                                                                        |            |           |
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